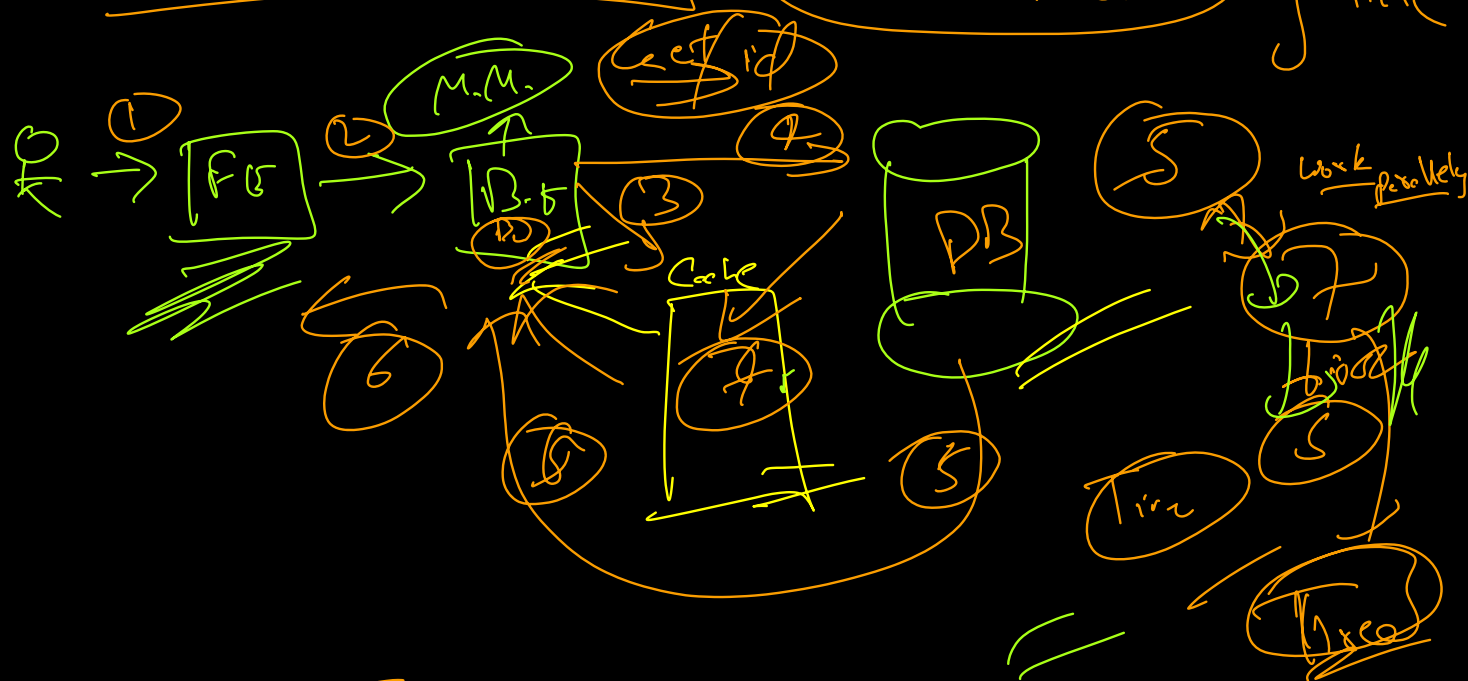
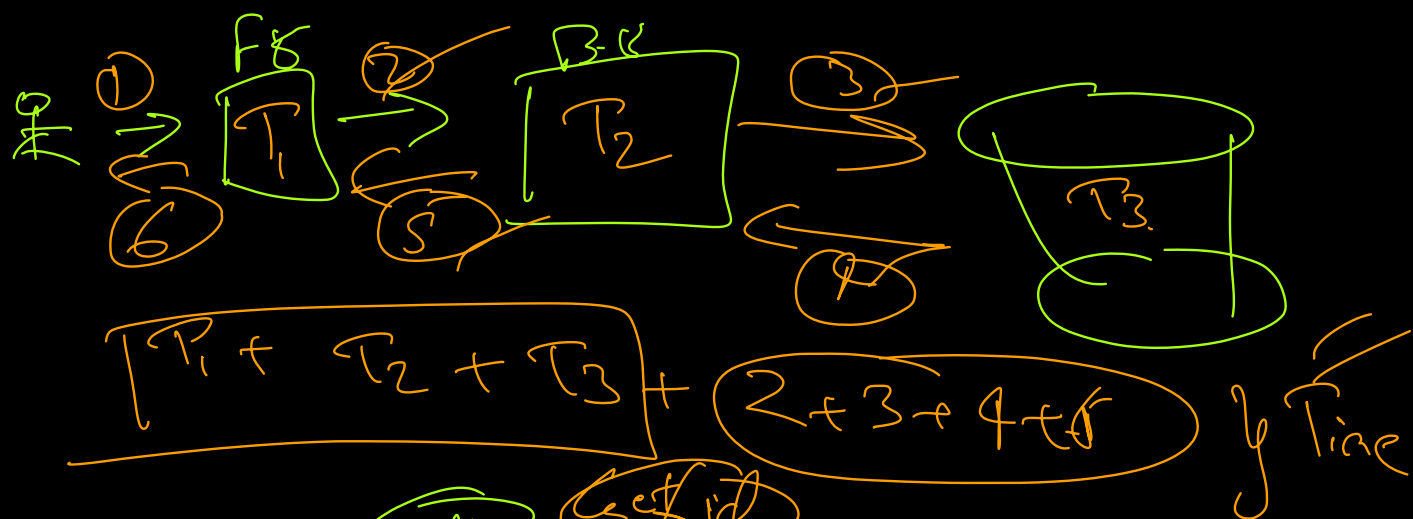
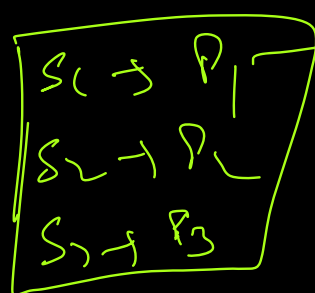




80% hit

20% Miss

Zero



expiry

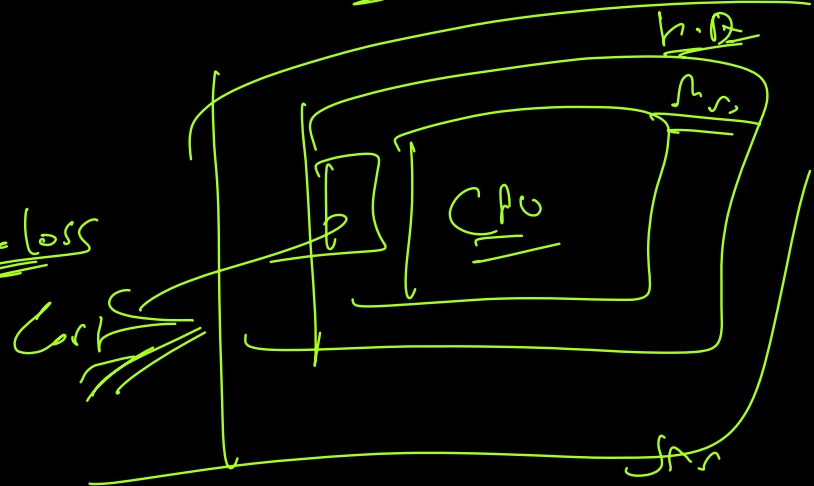
also keep a

Backup in H.D.

Pros

→ Time / Speed

→ Data Failure / Data Loss



Cons

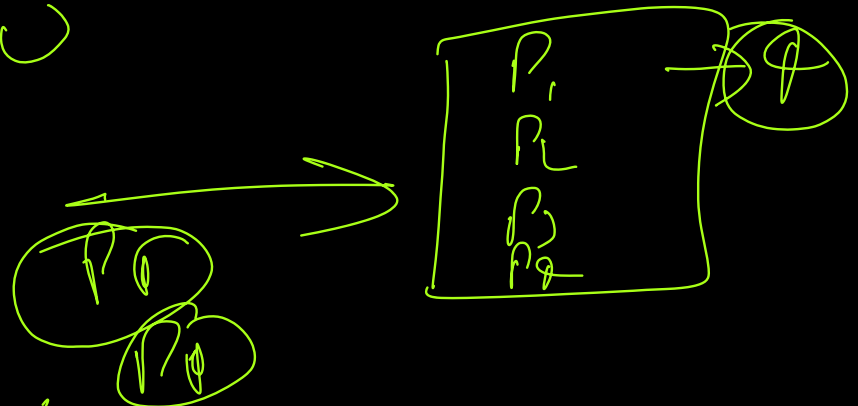
→ Time can be a problem if not implemented properly

→ Size of Cache & Clear Cache

Page Replacement Algor

(F.R.S)

L.R.U → Least Re U



1 2 3 4 3 1 2 1 3 4 3

4	3
3	4
2	5
1	2

(FIFO)
(LRU)

~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~2~~ ~~7~~ ~~1~~ ~~5~~ ~~3~~ ~~7~~ ~~7~~ ~~4~~ ~~3~~
 F.S = 4

(m.c)

~~n.d~~ ~~P~~
 1 1 1

P.A
 1 1 1
~~1 1 1~~

P.S = = F.C

~~M~~ ~~M~~ ~~M~~ ~~M~~ ~~n~~
~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~2~~ ~~3~~ ~~1~~

(2) ~~2~~ ~~7~~ ~~3~~ ~~1~~
 (3) ~~3~~ ~~7~~ ~~2~~ ~~1~~
 2 1 5 3

3 7 2 4
 3 4 2 7

8
 13

~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~2~~ ~~7~~ ~~5~~ ~~3~~ ~~7~~ ~~4~~ ~~3~~ (new)
 F.S = ~~4~~ 4

$\begin{bmatrix} 4 \\ 3 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 3 \\ 1 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 1 \\ 3 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 2 \\ 3 \\ 1 \end{bmatrix}$

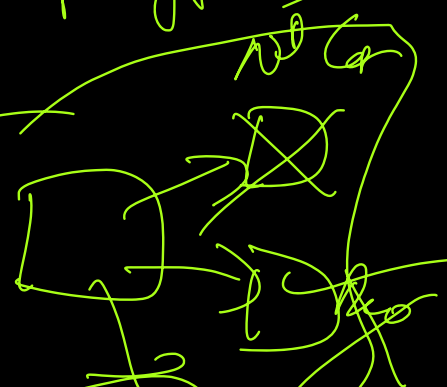
$\begin{bmatrix} 4 \\ 2 \\ 3 \\ 5 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 2 \\ 5 \\ 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 2 \\ 5 \\ 7 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 5 \\ 7 \\ 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 4 \\ 2 \\ 7 \\ 3 \end{bmatrix}$

in Realis

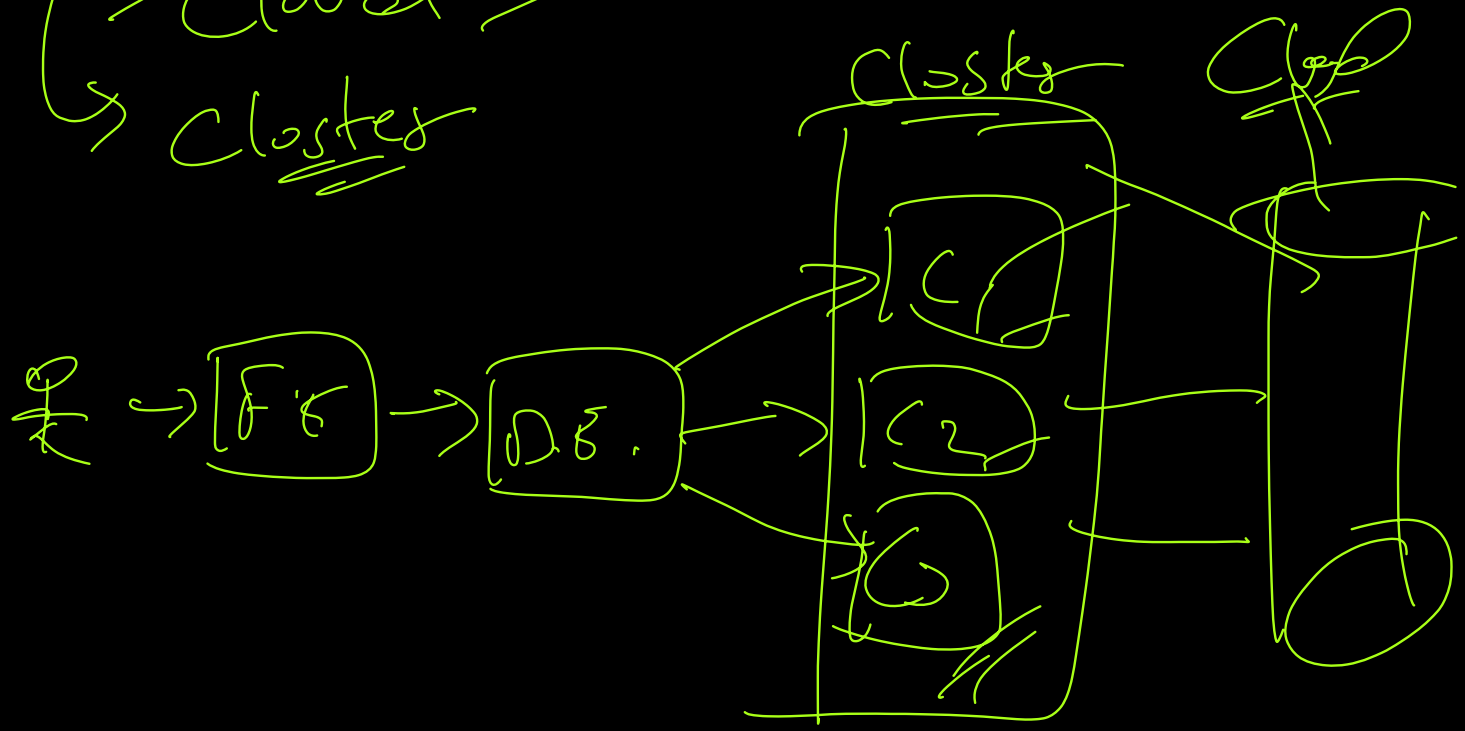
- ↳ Distributed Cache (avoid single pt. of failure)
- ↳ Replica on diff nodes
- ↳ Datatype

→ (String)

key



↳ local m/c
 ↳ cloud
 ↳ cluster



Data Structures

- ↳ (1) Strings
- ↳ (2) List (string)
- ↳ (3) Set (string)

$V \rightarrow V$

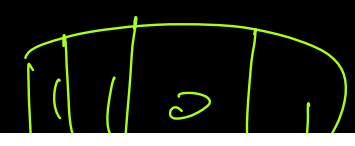
(4) Ordered set (string)

(5) Hash Map ($K \rightarrow \{F, V\}$)

(6) Geolocation

(7) Hyperloglog

(8) Bit



(8) 1300
(9) Stream

1300